

Name: Team Timbuktu

Date: 3/17/25

Element I: Testing, data collection, and analysis

In any engineering project, testing plays a crucial role in validating design concepts and ensuring both safety and functionality. Our class project involved the development of a toy enrichment hanger for an Eland, a large antelope requiring physical and mental stimulation in its environment. The primary goal of our design was to create a safe and engaging way for these animals to interact with toys, ultimately enhancing their well-being. Therefore, the importance of testing our design is significant; it confirms that our structure meets the necessary specifications and addresses any potential risks associated with its use.

Testing is a pivotal component of the engineering design process. For our project, we constructed two enrichment hangers, each standing 16 feet tall with a 10-foot arm that needed to extend over an 8-foot fence. This design ensured that the Eland could access the toys while remaining protected from external interference. Our initial tests focused on the strength and durability of the arm, which was intended to support toys weighing up to 100 pounds. To accurately assess the structural integrity of our design, we applied a simulated operational force of 80 pounds to the arm during our tests. This load provided a significant safety margin, as it was crucial to ensure that the structure could withstand unexpected pressures during its use. The testing environment was carefully controlled to mimic conditions the hanger would experience in actual scenarios, reflecting real interactions by the Elands.

The design analysis began by scrutinizing our initial prototype, which lacked support brackets. Connecting the arm directly to the vertical support structure left the joint vulnerable to failure under stress. During preliminary testing, we noted some flexing and movement at the joint, indicating a risk of structural failure that could harm the Elands. To enhance the integrity of the structure, we incorporated T brackets and L

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brackets into our design following the initial test. These reinforcements provided essential support at the connection between the arm and the vertical structure, effectively distributing forces encountered during use and significantly reducing the likelihood of joint failure. Subsequent tests after this modification showed considerably less flexing with the same applied force, demonstrating that our design revisions improved overall performance. In addition to assessing structural integrity, we also considered how the toys were positioned on the arm.

Our design allowed for variable placement, enhancing the interactive experience for the Elands while distributing weight more effectively. This multifaceted approach—addressing structural integrity, animal interaction, and safe operation—was crucial in creating a functional and safe enrichment device. The data implications from our testing provided invaluable insights into the effectiveness of our design revisions. Initial tests revealed that without reinforcements, our joint could not withstand the expected load, indicating an inadequate design for safe use. However, after integrating T and L brackets, we observed a marked improvement in the durability and stability of the structure during subsequent tests. Establishing a thorough testing regime was essential, as it demonstrated that our design could sustain the necessary forces while keeping the Elands safe. This iterative process affirmed the importance of conducting rigorous tests and making informed modifications based on those outcomes.

Moreover, the data collected highlighted the necessity for ongoing assessments and refinements in any engineering project. This experience not only deepened our understanding of the mechanics of design and testing but also reinforced the vital role of ensuring animal safety through responsible engineering practices. Our work on the toy enrichment hanger has enriched our knowledge of critically analyzing designs, engaging in thorough testing, and utilizing data to create safe and effective engineering solutions. The journey we undertook instilled a commitment to perseverance and self-advocacy in our engineering pursuits, which will undoubtedly serve us well in future projects.